Fiscal Year:	FY 2015	**Task Last Updated:**	FY 06/14/2015
PI Name:	Khusid, Boris Ph.D.		
Project Title:	Kinetics of Electric Field-Driven Phase Transitions in Polarized Colloids		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	07102-1982	**Congressional District:**	10
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	08/23/2013	**End Date:**	08/22/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nelson, Emily (NASA Glenn Research Center) Sheehan, Christopher (ZIN Technologies, Inc.) Loewen, Hartmut (University of Dusseldorf) Telo da Gama, Margarida (University of Lisbon) Yethiraj, Anand (Memorial University of Newfoundland)		
Grant/Contract No.:	NNX13AQ53G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Motivation: Colloidal assembly provides the capability for scalable manufacturing of structured materials and devices for traditional and emerging industries. Although equilibrium structures and thermodynamics of colloids exposed to a field have been well studied, the use of electrically assisted assembly is limited by a lack of understanding of the role of time and spatially varying fields in directing and controlling non-equilibrium processes. Microgravity offers an opportunity to study field-driven phenomena on non-buoyancy-matched colloids by removing masking gravity effects, such as particle sedimentation, convection, and jamming. Objectives: Design and conduct microgravity tests in the International Space Station Advanced Colloids Experiment facility in long timescales to elucidate fundamental microscopic mechanisms of non-equilibrium phenomena underlying the field-assisted assembly of interacting colloidal particles and suggest novel routes for creating functional materials. The emphasis will be on the field-driven evolution of the mesoscale phase transitions, electro-hydrodynamic instabilities, and nucleation and growth of crystalline structures. The phenomena selected are logical candidates to take advantage of microgravity because terrestrial observations have failed to provide a comprehensive physical understanding of the underlying physics. Methodology: Dielectrophoretic experiments will be carried out on well-characterized colloids under well-defined conditions in terrestrial and microgravity environments. We will use electro-hydrodynamic methods and apparatuses for the control and manipulation of particles which were developed in our NASA-supported projects, including parabolic flight tests conducted in 2002 and planned in the summer of 2013. Molecular dynamics simulations, coarse-grained and continuum models will be employed to study field-induced colloidal structures. Deliverables: Detailed understanding of electrically driven non-equilibrium phenomena in colloids, apparatuses for the field-assisted colloid assembly, experimental database, and predictive models for the field-induced structure. The research will be integrated with educational and outreach initiatives in space sciences and related technologies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research has the strong potential to reveal the fundamental microscopic mechanisms of electrically driven phenomena in a suspension of interacting particles and develop a physical explanation for puzzling observations under terrestrial conditions that are not predicted by current theories. The main focus is on the appearance of a cellular structure in a colloid exposed to a spatially uniform field and understanding of the nucleation and growth of crystalline structures in a high-gradient and mechanisms of electro-hydrodynamic instabilities of concentration fronts. The obtained results will facilitate the development of new technologies for the control and manipulation of colloidal processes. Space applications: Electrical technologies can be employed in space to mitigate the effects of reduced gravity by providing an electrical body force. Results of ACE-T-E experiments will be useful for the development of new methods for electro-hydrodynamic control, manipulation, and separation of multiphase fluid flows in various space applications, for example, thermal management systems. Earth applications: The research should bring the understanding of electrically driven colloidal processes to the level useful in the design of apparatuses and processes for a broad range of applications in materials engineering, life science, electronics, and chemical industry. This will stimulate further "spin-offs" of electrical technologies.
	Major Accomplishments The study was conducted within the scope of the originally proposed research plan. The New Jersey Institute of Technology (NJIT) team worked in closed collaboration with NASA scientists preparing International Space Station Advanced Colloids Experiment experiments. The main focus was on the development requirements for a high-voltage amplifier and an optical system. The NJIT team carried out ground-based experiments needed to specify basic parameters for the design of the dielectrophoretic cell geometry, electrode configuration, voltage/frequency regimes, and preparing suspensions for the ISS experiments. The PI participated in workshops of European Space Agency (ESA) Colloidal Solids Science teams working on the development and tests of the dielectrophoretic technology for ISS experiments (virtually at European Space Research and Technology Centre (ESTEC), Noordwijk, the Netherlands, July 1 and November 6, 2014 and face-to-face at Qinetiq Space, Kruibeke, Belgium, February 25-26, 2015). The PI is also involved in establishing research collaboration between NASA and Russian Federal Space Agency -ROSCOSMOS to study dielectrophoretic phenomena in suspensions under microgravity in the ISS. Hardware: A setup to be developed for experiments in the ISS should provide AC high-voltage to energize electrodes of a dielectrophoretic cell and control the specimen temperature. An optical system will be used for imaging the field-driven evolution of the spatial distribution of particles in the cell and collecting small angle light scattering (SALS) patterns, when experiments are conducted in the ESA Colloid Light Scattering (COLIS) instrument, to provide structural information about the particle distribution. Ground based experiments conducted by NJIT team demonstrated that requirements for the setup can be modified: • A field frequency range in the ISS experiments can be reduced from MHz to 60 kHz and the gap of a dielectrophoretic cell can be increased from 0.1 mm to about 0.5-0.7 mm, • Electrodes can be placed outside of the dielectrophoretic cell, • Image size of a camera for imaging particle structures formed during the exposure to an electric field should be at least 2 x 2 mm that appears to be compatible with the current design of an optical system for the ESA COLIS instrument. Advantages of the proposed features: • Ease of fabrication of the ISS hardware and stirring specimens in the ISS experiments, • Drastic reduction of a relative variation of the gap thickness across the cell that would reduce undesirable field gradients within the cell, • Increase of the flow Reynolds number that would facilitate homogenization of a specimen before and after exposure to an electric field, • Eliminating causes of electrochemical corrosion of electrodes, • Drastic reduction in the possibility of forming bubbles inside a specimen during the exposure to an electric field that would destroy the specimen, • Reduction of ambiguity in interpretation of turbidity and SALS measurements in the ESA COLIS instrument.

Task Progress:	Materials: Experiments in microgravity will be conducted on dilute and relatively concentrated suspensions (0.5-10 v/v %, 1-15 micrometers) of non-buoyancy-matched and buoyancy-matched polarized particles. Buoyancy-matched suspensions would serve as “control samples” for comparing results of ground-based and microgravity tests. We expect to conduct experiments on several types of particles and several concentrations for each type of particles to cover a wide range of the mismatch of dielectric constants and conductivities between particles and a suspending liquid. NJIT team conducted ground-based experiments to specify the particle type, the suspending liquid, and the number and volume of specimens of different concentrations within each type may change, as specialty cell designs, high-voltage unit. Preliminary choice of suspensions: Commercially available particles of aluminum, polyethylene, and polyalphaolefin suspended in corn oil with 1-5 % of commercially available dispersant/stabilizer based on Guerbet alcohols. Operational requirements: A dielectrophoretic cell for ISS experiments will be fully filled with a suspension in the Earth facility. As the particles will sediment in Earth’s gravity during storage and the hyper-gravity during the spacecraft launch, specimens should be homogenized prior to experiments. A homogenized specimen will be heated to a desired temperature and then exposed to an electrical field by applying a desired voltage/frequency for a certain period of time. The optical system will provide images of the field-driven evolution of the suspension structure and collect SALS patterns when experiments are conducted in the ESA COLIS instrument. To quantify effects of electric fields on colloidal processes, experiments should be carried out on each specimen at different voltages and frequencies and, if possible at several different temperatures. After exposure to an electric field, a specimen should be homogenized prior to another test. For each suspension, 3 - 6 magnitudes of applied voltage and several frequencies at each voltage will be tested for each selected temperature. Several tests at selected voltage/frequency regimes will be repeated for evaluating reproducibility of microgravity experiments. The voltage/frequency regime and duration of a particular microgravity experiment will be determined in real-time depending on the previous results as data sets are transmitted to Earth. These parameters cannot be predicted based on observations of the field-driven processes in non-buoyancy-matched suspensions under Earth’s gravity. Two techniques are currently considered for homogenization of a suspension after the exposure to an electric field prior to another test. One of them is to use a peristaltic pump to flow the exposed specimen from the dielectrophoretic cell to the container and simultaneously the fresh specimen from the container into the cell. The other is to employ electro-hydrodynamic flow generated in the dielectrophoretic cell by applying a low frequency electric field. It would allow conducting several tests on each specimen. This technique was successfully demonstrated in ground-based experiments.
Bibliography Type:	Description: (Last Updated: 02/06/2025)
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